



UNIVERSITY OF
GOTHENBURG

Workshop 2: EDA

VASIL MANKEVICH, PHD

- Exploratory Data Analysis — a systematic approach to
- investigating datasets before modeling or testing hypotheses.
- "The greatest value of a picture is when it forces us to notice what we never expected to see."
- Goals:
 - Familiarity — understand structure, types, size
 - Quality — find errors, missing values, outliers
 - Patterns — distributions, correlations, groups
 - Storytelling — communicate findings clearly

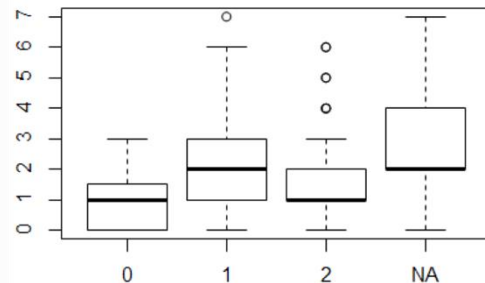
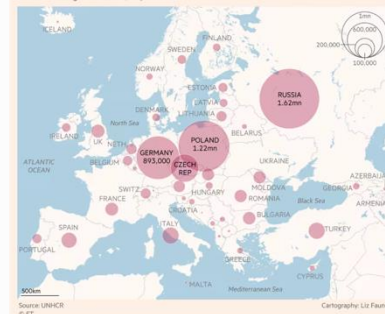
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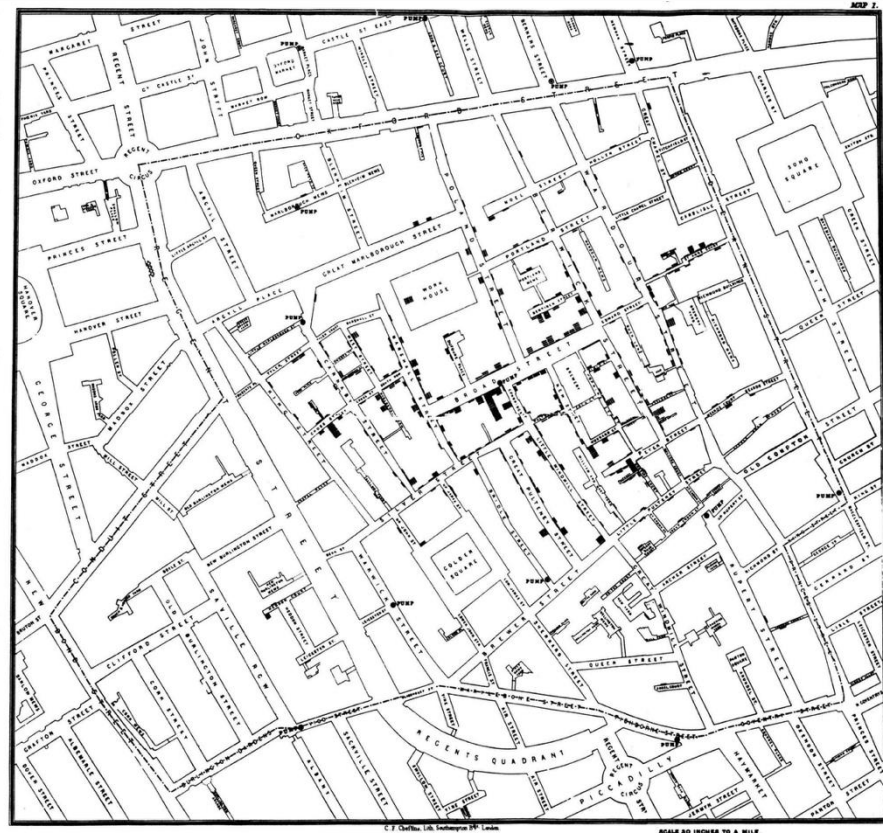
Ukrainian refugees seek safety in multiple countries

Estimated refugees recorded, July 12 2022



Exploratory tasks

- 1854 Broad Street Cholera Outbreak Map* – one of the first data visualizations
- **Summarize, visualize, diagnose, communicate**
- Is Excel enough?
- Tableau, QlikView, Microfof BI?



- Exploratory data analysis is always needed
- Visualization can be great for exploration and communication





Type	Abbr.	Meaning	Example
Missing Completely At Random	MCAR	Missingness unrelated to any variable	Sensor randomly fails
Missing At Random	MAR	Missingness depends on observed variables	Casual users skip surveys
Missing Not At Random	MNAR	Missingness depends on the missing value itself	Older people hide their age

Enders, C. K. (2022). *Applied missing data analysis*. Guilford Publications

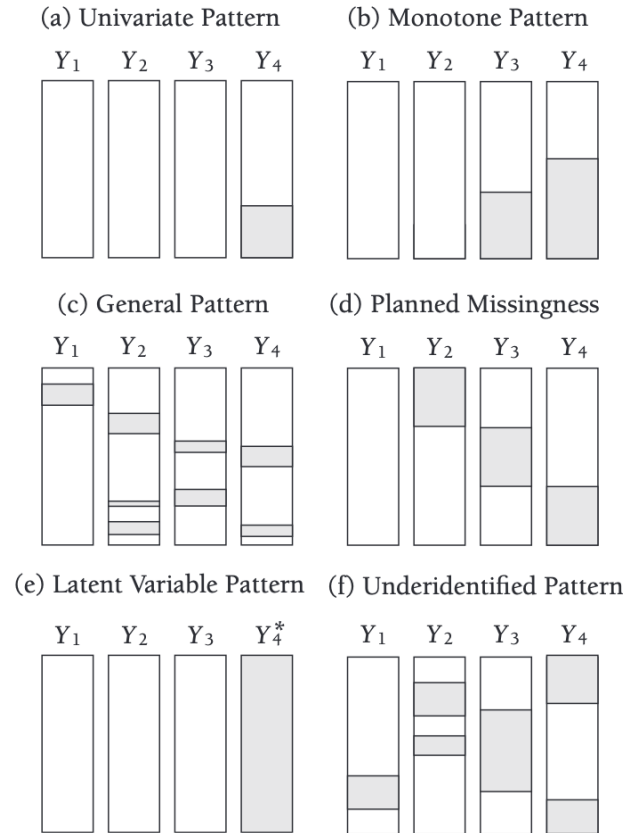
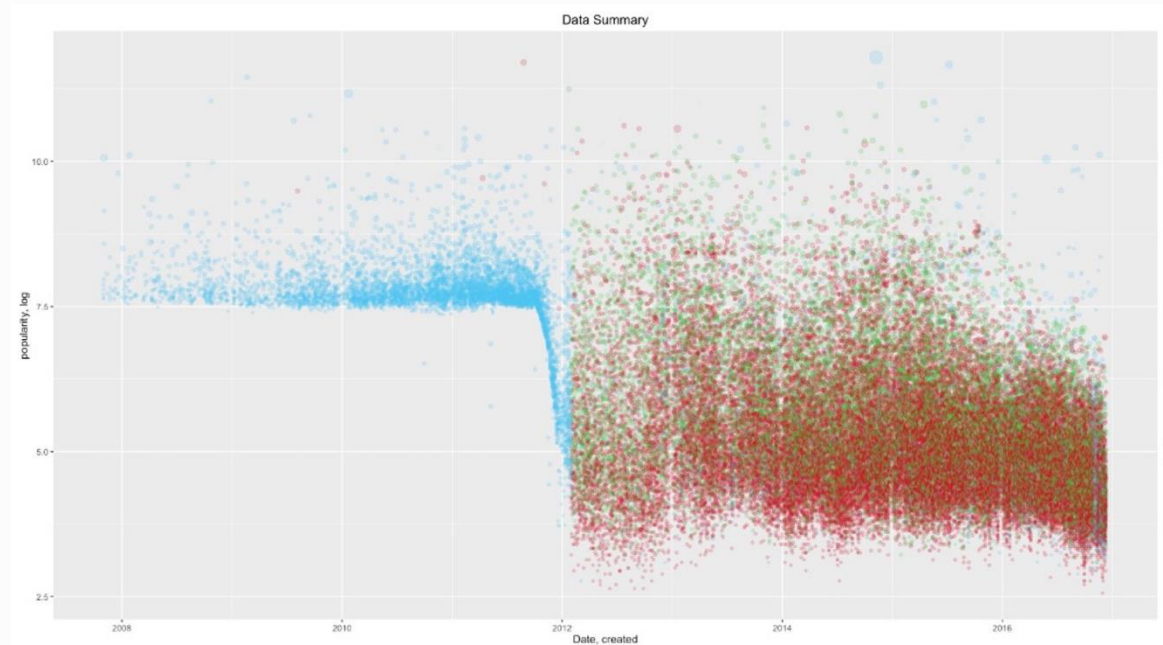
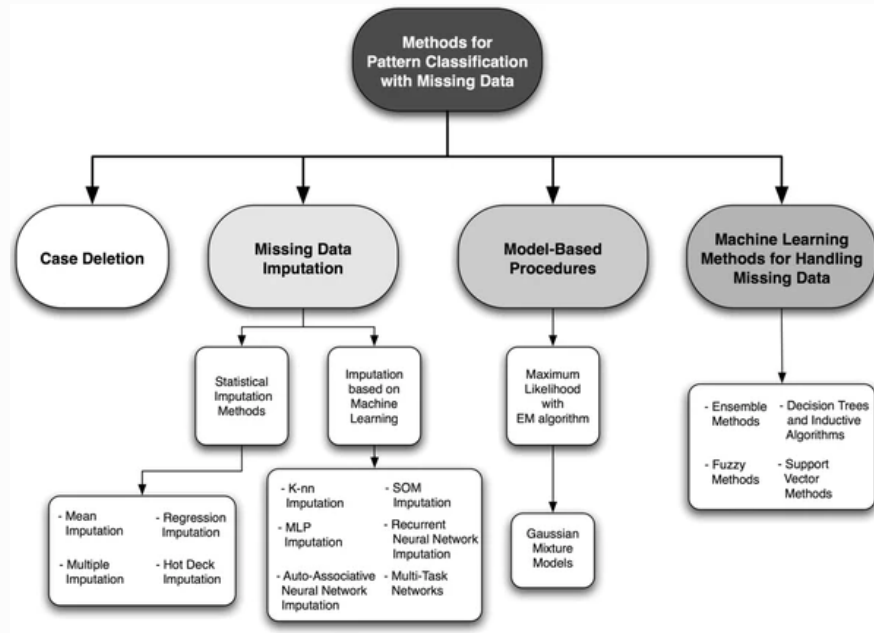


FIGURE 1.1. Six missing data patterns. The gray shaded areas of each bar represent missing observations.

- Patterns of missing data: at random, completely at random, not at random
- How to deal with NAs:
 - Deletion: Columns, rows
 - Imputation: special values, means, KNN etc.





García-Laencina, P.J., Sancho-Gómez, J.L. & Figueiras-Vidal, A.R. Pattern classification with missing data: a review. *Neural Comput & Applic* 19, 263–282 (2010).
<https://doi.org/10.1007/s00521-009-0295-6>

Exploratory analysis toolbox

- Categorical variables: frequency – barplot
- Numerical variables: stats and histograms
- Boxplots
- Outliers
- Dispersion, center, range, variance
- Count averages, sums
- Missing values

Some approaches for text

- Missing values
- Length
- Readability
- Unique terms
- Frequency
- Language
 - Which language is the text?
 - Mixed languages
- Keywords-in-context